MODULE III : Composites

(No. of hours – 04, Weightage of marks with option – 14)

- **Course Objective**: To learn the composition, properties and functions of various Composite Materials.
- **Course Outcome**: To identify different types of Composite Materials for the industrial uses.
- **Prerequisite**: Knowledge about Constituents of Composites and their functions.

MODULE III : Composites

Sr. No.	TOPIC	REMARKS
1.	Types of Composites, Applications i)Fibre-reinforced composites- e.g. Glass fibre composites, Carbon fibre reinforced composites, Aramid fibre reinforced composites ii)Layered composites (Laminates) iii)Particulate composites- large particle and dispersion strengthened composites	Introduction to composites Classification of Composites needs to be discussed. Description and applications of fibre-reinforced composites. Structural composites; laminates and sandwich panels. Description and application of only laminates. Description of particulate composites and their applications. No questions will be asked on classification.
2.	Bio-composites— Definition, Classification, Applications	What are bio-composites? Based on partial bio-degradable and completely bio-degradable bio-composites.

Reference Books (Composites):

1. Engineering Chemistry - Jain and Jain, Dhanpat Rai Publication.
2. A textbook of Engineering Chemistry by Shashi Chawla, Dhanpat Rai Publications.
3. Material science and Metallurgy, O. P. Khanna, Dhanpat Rai Publications.
4. Bio-composites- Recent Advances, Edited by Magdy M.M. Elnashar and Selcan Karakuş Intechopen

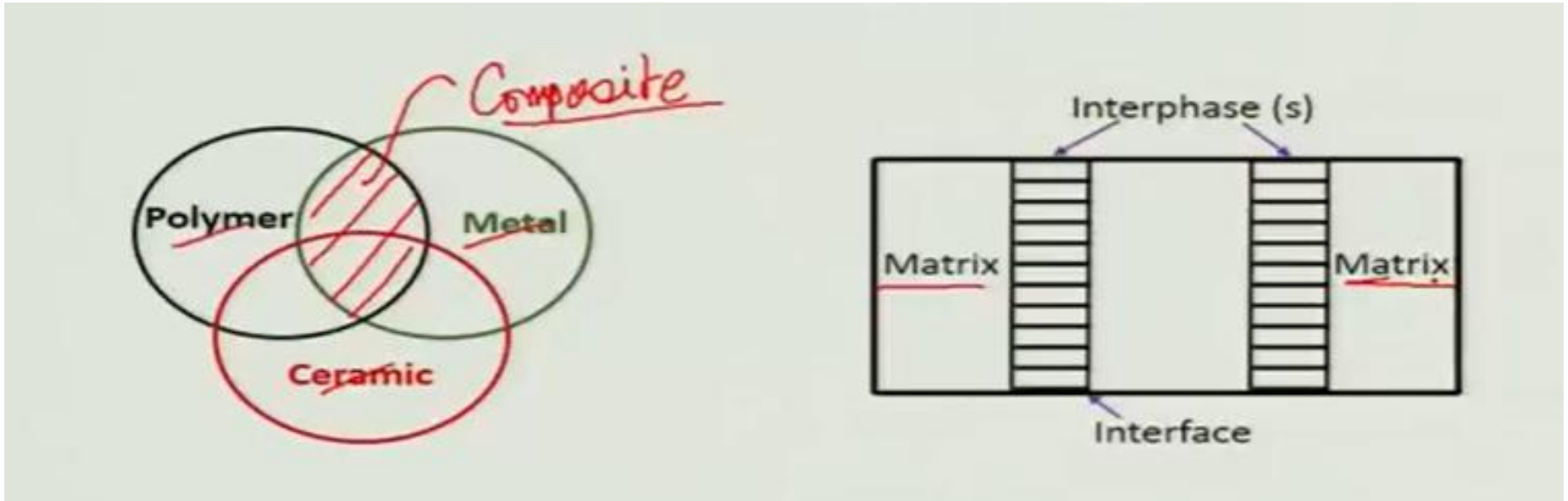
What is a composite Material?

Definition: Composite materials are engineering materials made from two or more constituent with significantly different chemical properties and physical properties which when combined macroscopically, produce a new substance with different properties from the individual component.

Examples: 1. Fiberglass: Made from the glass fiber woven into a cloth and bonded together with a resin. It is used in ship hulls, car bodies storage tanks etc.

2. Carbon Fiber Reinforced Polymer(CFRP): Composed of carbon fibers embedded in a polymer matrix. It is used in aerospace, sports equipment etc.

3. Concrete: A mixture of cement , water ,sand and gravel. It is widely used in concentration due to its compressive strength and durability.


Composite
Polymer
Metal
Ceramic
Interphase (s)
Matrix
Matrix
Interface

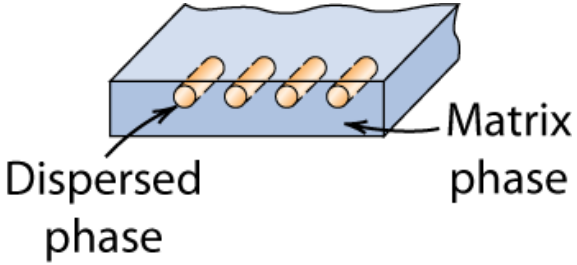
Components in a Composite Material

Nearly all composite materials consist of two phases:

1. Primary phase - forms the *matrix* within which the secondary phase is imbedded. Matrix Phase is continuous phase

2. Secondary phase - Imbedded phase sometimes referred to as a reinforcing agent/ reinforcing phase / Dispersed Phase, because it usually serves to strengthen the composite. Dispersed phase is discontinuous and surrounded by matrix phase

The reinforcing phase / Dispersed Phase may be in the form of fibers, particles, or various other geometries.



Each plays a crucial role in determining the properties of the composite

Types of Matrix Phase:

- **Polymer Matrix Composite(PMC):** It includes thermosetting (such as epoxy, polyester)and thermoplastic (such as polyethylene, polypropylene).Polymer matrix is light in weight, corrosion resistant and relatively easy to process.
- **Metal Matrix Composite(MMC):** It includes metals like aluminum and titanium as the matrix. It offers higher temperature resistance , wear resistance and higher strength compared to polymer matrix.
- **Ceramic Matrix Composite(CMC):** It includes ceramics like silicon carbide and aluminum oxide as the matrix phase. It offers thermal stability, corrosion resistance and high strength at elevated temperature.

Classification of dispersed phase

- The dispersed phase can be fibre particle etc.
- Fibres:

1.Glass fibres

2.Carbon fibres

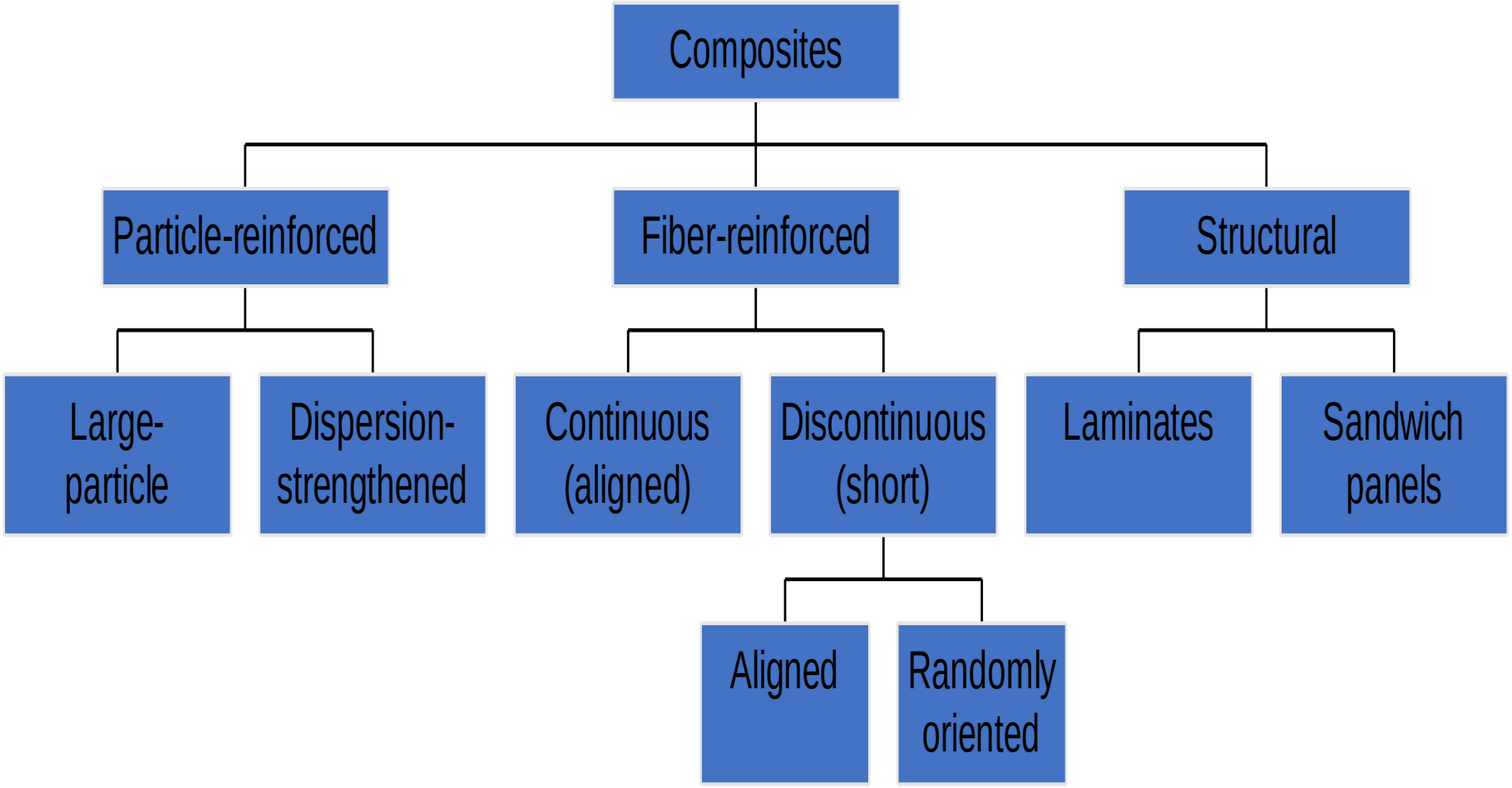
3. Aramid fibres

Dispersed phases – functions and applications

Type of Dispersed Phase	Function	Example	Applications
A. Fibres			
1. Glass Fibres: obtained by forcing molten glass through spinneret	Provides tensile strength, chemical resistance and of low cost.	Glass fibre reinforced polymer.	Boat hulls, car bodies.
2. Carbon Fibres: obtained by pyrolysis of organic fibres in inert atmosphere	High strength to weight ratio, stiffness.	Carbon fibre reinforced polymer.	Aerospace, sports equipment, high-performance vehicles
3. Aramid Fibres: is made by spinning liquid crystal aramid oligomers (Kevlar).	These are high performance fibres. (High modulus and tensile strength, stability at high temps, resistant to creep and fatigue failure).	Armid fibre reinforced polymer.	Aramid fibres are use along with polyesters or epoxides.

Composite	Fibre Phase	Matrix phase	Properties	Application	Limitation
Glass Fibre reinforced Polymer Composite	Glass fibres	Polymers such as nylons , polymers	Lower density, Higher tensile strength, Higher impact resistance, high corrosion resistance	Automobile parts, Transportation Industries, storage tanks	Decomposed at low temperature
Carbon Fibre reinforced polymer composite	Carbon	Polymers	Lighter, Excellent resistance to corrosion, Stability at high temperature	Structural Components e.g. wings of aircraft helicopter, fishing rods	High cost
Aramid Fibre reinforced composite	Short fibre	Polymers	High surface area, Toughness to strength Heat stability Wear Resistance	Brakes , Clutches	Price is high

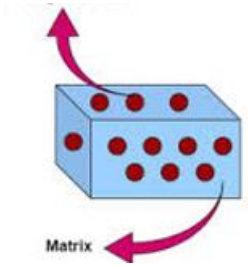
Classification of Composites



Particle reinforced Composites

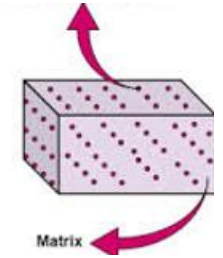
Large-particle RC

- LPRC have relatively large particles dispersed throughout the matrix material. (mm or micrometer range)
- Concrete – used in construction, Metal-matrix composites – used in automotive and aerospace industries
- Improve strength, stiffness, resistance to wear and tear



Dispersion-strengthened RC

- DSRC are very fine particles, typically in nm to micrometer range, dispersed throughout the matrix material.
- Ceramic composites – used in high temperature applications like turbine blades
- Improved strength and toughness, stiffness, creep resistance



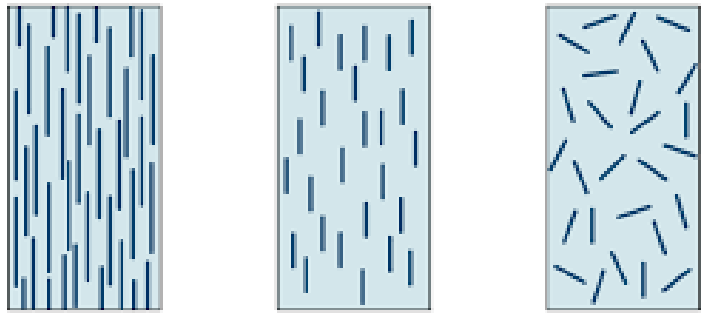
Fibre reinforced Composites

Continuous fibre RC – (long) aligned

- CFRC use long, unbroken fibres, aligned in a specific direction within the matrix
- High strength, stiffness, fatigue resistance
- Applications in aerospace, automotive, and sports equipments – tennis rackets, fishing rods

Discontinuous fibre RC – (short) aligned and randomly oriented

- DFRC use short aligned or randomly oriented fibres that do not extend across the entire composite
- Improved strength and stiffness, directional strength, ease of processing
- Applications in Construction, automotive and consumer goods



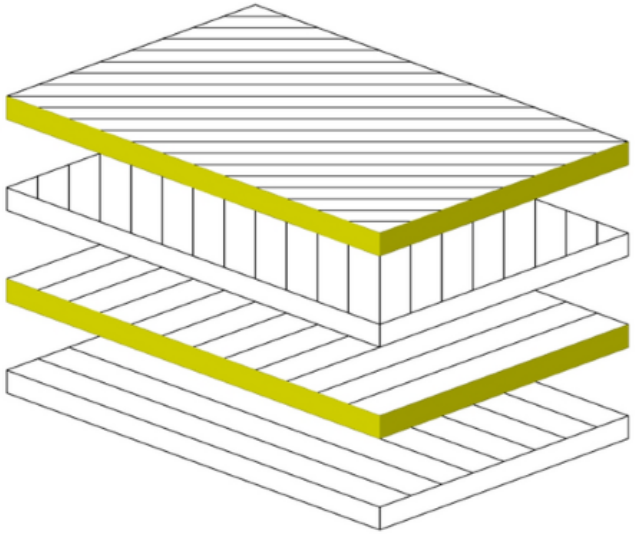
Structural composites

- The materials made from two or more distinct components, that, when combined, create a material with superior properties.
- Example materials:

- Sandwich panels (Honeycomb panel)
- Laminates (e.g. plywood)

Laminated composites

- They are made by stacking and bonding together multiple layers of materials
- Each layer, or ply, may have a different orientation or material properties
- Layers of composite materials are stacked and bonded together to form a strong, lightweight structure
- For example – In Plywood, the laminates are thin sheets of wood staked in alternating grain directions and bonded with adhesive
- This alternating grain orientation enhances the overall strength and stability of the panel

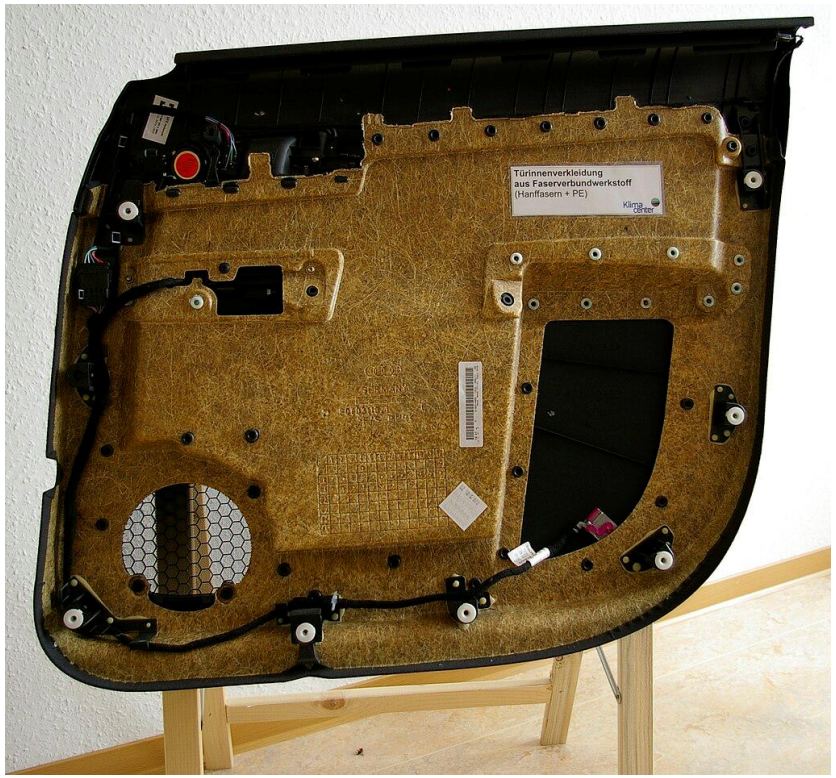


- Applications – furniture, aerospace, marine, satellite components, protective gear (helmets), armored vehicles, etc.

Biocomposites

- A biocomposite is a composite material formed by a matrix (resin) and a reinforcement of natural fibers (bio fibres).
- The matrix phase is formed by polymers derived from renewable and nonrenewable resources.
- Biofibers are the principal components of biocomposites, which are derived from biological origins, for example fibers from crops (cotton, flax or **hemp**), recycled wood, waste paper, crop processing byproducts or regenerated cellulose fiber (viscose/rayon).

Biocomposites



Interior carpeting of a car's door made by a biocomposite of hemp fibres and polyethylene

Classification of Biocomposites

- Biocomposites can be classified into two main categories based on their biodegradability:

1. Partially Biodegradable Biocomposites-

- Made from a combination of biodegradable natural fibers (e.g., plant fibers, wood fibers) and non-biodegradable synthetic polymers (e.g., polypropylene, polyethylene).
- The natural fibers provide biodegradability, while the synthetic polymers provide durability and strength.
- Examples: Wood-plastic composites, plant fiber-reinforced polypropylene

Classification of Biocomposites

2. Completely Biodegradable Biocomposites-

- Made entirely from biodegradable materials, including natural fibers (e.g., plant fibers, wood fibers) and biodegradable polymers (e.g., polylactic acid, polyhydroxyalkanoates).
- These biocomposites can easily decompose and return to nature.
- Examples: Plant fiber-reinforced polylactic acid, bacterial cellulose-reinforced starch-based bioplastics.

Applications of Biocomposites

1. Packaging: Biodegradable packaging materials, such as containers, bags, and wraps.

2. Automotive: Bio composite materials for car parts, like dashboards, door panels, and seat components.

3. Aerospace: Lightweight biocomposites for aircraft and spacecraft components.

4. Construction: Biocomposite materials for building insulation, roofing, and wall panels.

5. Furniture: Biodegradable furniture made from biocomposites, such as chairs, tables, and cabinets.

Applications of Biocomposites

6. Textiles: Biocomposite fibers for clothing, upholstery, and technical textiles.

7. Medical: Biocompatible materials for medical implants, surgical instruments, and wound dressings.

8. Consumer Products: Biodegradable products, such as disposable cutlery, plates, and cups.

9. Agricultural: Biocomposites for agricultural applications, like biodegradable mulch films and plant pots.

10. Sports Equipment: Lightweight biocomposites for sports equipment, such as bicycle frames, golf clubs, and ski boots.